

Muhammad Fakhrrur Rozi

+62 851 1130 4115 | Central Java, Indonesia | marrazy54@gmail.com | github.com/LitFill | linkedin.com/in/muhammad-ar-razy-b04179208/ | rozy.my.id

SUMMARY

Self-taught in computer science during and after an Arabic Literature degree, driven by a fascination with formal languages and type systems. Functional programmer with deep experience in Haskell and Koka, actively learning Scala 3. Passionate about type systems, compilers, language design, category theory, and purely functional design. Seeking a junior role where I can contribute to production FP systems and grow alongside a strong engineering team.

EDUCATION

Ma'had Aly Andalusia Banyumas Banyumas, Indonesia
Sarjana Sastra Arab (Arabic Literature) 2021 — 2025

- GPA: 3.65 / 4.0
- Self-taught in CS during and after this degree, driven by fascination with formal languages and type systems

SMA Islam Andalusia Kebasen Banyumas, Indonesia
High School 2017 — 2020

SKILLS

- **Primary Languages:** Haskell, Koka, OCaml, Rust, TypeScript/JavaScript
- **Familiar With:** Go, Idris2, Agda, Lean, Common Lisp, Nushell
- **Haskell Ecosystem:** servant, warp, aeson, megaparsec, lens, QuickCheck, hedgehog, streaming, polysemy, hspec, tasty, req, hasql, relude
- **Tools:** Git, Linux (NixOS, Arch), Neovim, Cabal, GHCup, HLS, Stack, Nix, hledger, Node.js, Playwright
- **Soft Skills:** Self-directed learning (15+ languages independently), technical communication, teaching & mentoring

PROJECTS

derive — Koka Derive CLI <https://github.com/LitFill/derive>
• CLI tool eliminating boilerplate by auto-deriving show, eq, and typeclass-like functions in Koka (2 derivation kinds, file I/O, dry-run mode). Built because Koka lacked metaprogramming — solved a real workflow pain point.

sina — Hindley-Milner Type Inference <https://github.com/LitFill/sina>
• Full HM type inference (Algorithm W) from scratch in Haskell: parser, AST, constraint-based type checker with let-polymorphism. Supports integers, booleans, conditionals, let-bindings, binary ops.

klens — Value-based Lenses in Koka <https://github.com/LitFill/klens>
• Value-based lenses for Koka: one-shot focused views with set, effectful modify propagating Koka effects, composition, and tuple optics — a design exercising Koka's effect system.

src-todo — TODO Tracker CLI <https://github.com/LitFill/src-todo>
• Haskell CLI managing TODO comments in source files: register, list, and unregister todos with unique UUIDs, updated in place. Comment parsing built with Megaparsec.

EXPERIENCE

Secretary / Administrative Automation 2020 — Present
Yayasan Al Anwar Al Hisyamiyyah — Pondok Pesantren Banyumas, Indonesia

- Built Playwright automation scripts to batch-enter student data for 3,000+ santri into the EMIS (Education Management Information System), eliminating manual data entry entirely
- Developed Node.js docx generators and OOXML manipulation tools to produce Laporan Pertanggungjawaban (accountability reports) for Pondok management, reducing preparation time from days to minutes
- Designed document automation workflows combining Playwright, OOXML, and custom scripts to streamline administrative reporting across 10+ report types

Programming Instructor Jan 2026 — Jun 2026
Ma'had Aly Andalusia (Extracurricular) Banyumas, Indonesia

- Taught programming fundamentals to students with no prior coding experience — 10+ students over 6 months
- Designed curriculum and hands-on exercises from scratch, adapting content for absolute beginners
- Developed teaching materials and guided students through practical coding sessions

OPEN SOURCE

- **BNFC** — Reported and fixed extraneous closing bracket in string literal doc generation
- **Noctalia Shell** — Custom plugins for the Noctalia desktop environment

CERTIFICATIONS

- English Language Certification — BLK, Kementrian Ketenagakerjaan

LANGUAGES

- English (professional working), Indonesian (native), Arabic (academic)